

Write-up.

Practice proving valid arguments.

Show each argument is valid by listing the steps and the rules of deduction as we did in class.

1. $A \rightarrow C, A \wedge B \therefore C$

1. $A \wedge B$ Premise
2. A Conjunctive Simp. ①
3. $A \rightarrow C$ Premise
4. C Modus Ponens 2, 3

2. $(B \vee C) \rightarrow A, B \therefore A$

1. B Premise
2. $B \vee C$ Disjunctive addition ①
3. $(B \vee C) \rightarrow A$ premise
4. A Modus Ponens 2, 3

3. $A \vee B, B \rightarrow D, \sim C, D \rightarrow C \therefore A$

1. $D \rightarrow C$ Premise
2. $\sim C$ Premise
3. $\sim D$ Modus tollens 1, 2
4. $B \rightarrow D$ Premise
5. $\sim B$ Modus tollens 3, 4
6. $A \vee B$ Premise
7. A Disjunctive Syllogism 5, 6

4. $A \leftrightarrow B, A \vee C, \sim B \therefore C$

1. $A \leftrightarrow B$ premise
2. $(A \rightarrow B) \wedge (B \rightarrow A)$ Logical Equivalence ①
3. $A \rightarrow B$ Conjunctive simp. ②
4. $\sim B$ Premise
5. $\sim A$ Modus tollens 3, 4
6. $A \vee C$ Premise
7. C Disjunctive Syllogism 5, 6

Over $\rightarrow$

$$5. A \vee C, B \vee D, \sim(A \vee B) \therefore C \wedge D$$

1. $\sim(A \vee B)$ Premise
2. $(\sim A) \wedge (\sim B)$ DeMorgan ①
3. $\sim A$ Conj. Simp ②
4. $A \vee C$ Premise
5. C Disj. Syllogism 3, 4
6. $B \vee D$ Premise
7. $\sim B$ Conj. Simp ②
8. D Disj. Syllogism 6, 7
9. $C \wedge D$ Conj. Addition 5, 8

$$6. (\sim A) \rightarrow B, B \rightarrow C, (\sim A) \vee (\sim D), D \therefore C$$

1. D Premise
2. $(\sim A) \vee (\sim D)$ Premise
3. $\sim(\sim D)$ Double negation ①
4. $\sim A$ Disj. syllogism 2, 3
5. $(\sim A) \rightarrow B$ Premise
6. B Modus ponens 4, 5
7. $B \rightarrow C$ Premise
8. C Modus Ponens 6, 7

Exercise your creativity! Make up your own valid argument with at least 4 variables and at least 7 steps. Maybe yours will be chosen as an exam question.

For extra practice, use the arguments given in exercises 33-42 in the book but cover up the steps.